

Parallel and Distributed Computing

Lecture 7

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✧ Distributed systems are defined as a collection of multiple independent systems connected together as a single system. Every independent system has its own memory and resources and some common resources and peripheral devices that are common to devices connected together. The design of Distributed systems is a complex process where all the nodes or devices need to be connected together even if they are located at long distances. Challenges faced by distributed systems are Fault Tolerance, transparency, and communication primitives. Fault Tolerance is one of the major challenges faced by distributed systems.

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- ✧ In distributed systems, there are three types of problems that occur. All these three types of problems are related.
 - ✧ **Fault:** Fault is defined as a weakness or shortcoming in the system or any hardware and software component. The presence of fault can lead to error and failure.
 - ✧ **Errors:** Errors are incorrect results due to the presence of faults.
 - ✧ **Failure:** Failure is the final outcome where the assigned goal is not achieved.

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- ✧ Fault Tolerance is defined as the ability of the system to function properly even in the presence of any failure. Distributed systems consist of multiple components due to which there is a high risk of faults occurring. Due to the presence of faults, the overall performance may degrade.

✧ Fault tolerance is the ability of a system to continue functioning correctly even when some of its components fail. In distributed systems, fault tolerance is essential for ensuring **reliability**, **availability**, and **consistency**.

✧ **Key Goals**

1. **Reliability**: The system performs its intended function correctly.
2. **Availability**: The system is operational and accessible when needed.
3. **Consistency**: The system maintains correct and consistent data across nodes.

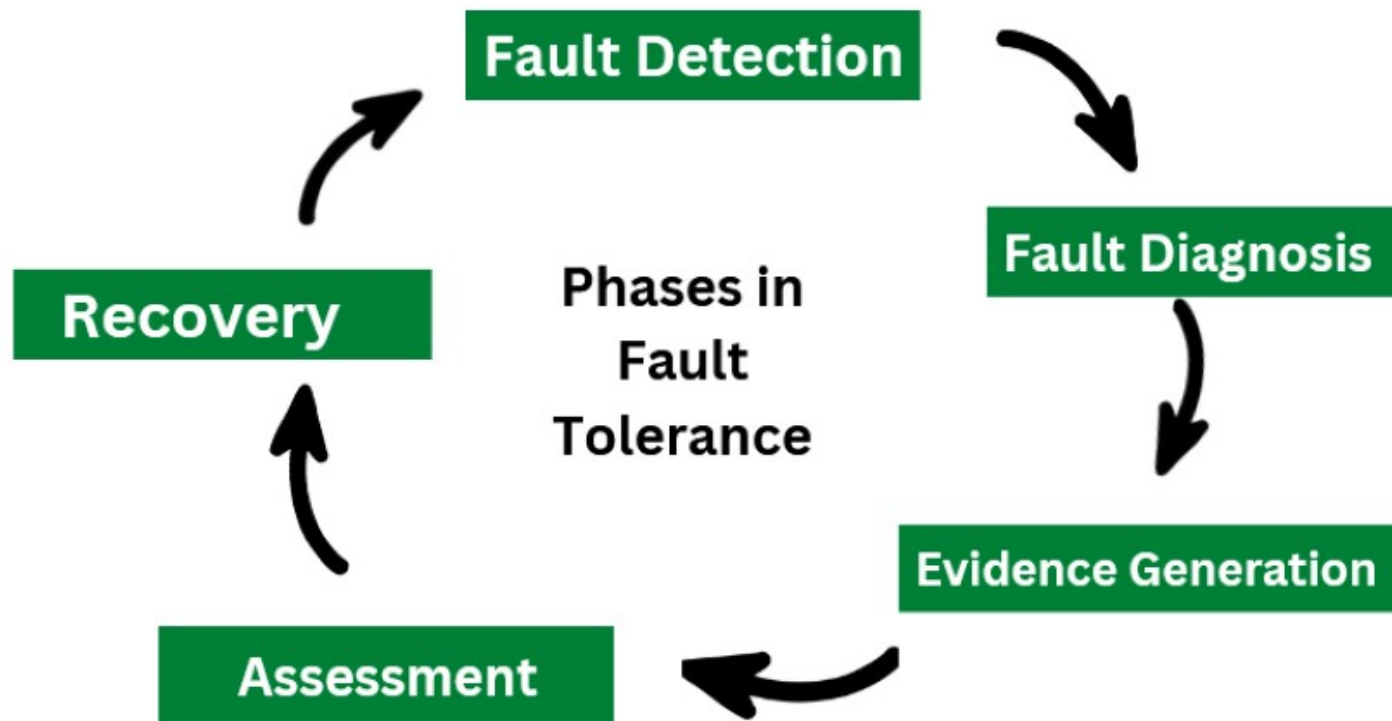
✧ Techniques for Fault Tolerance

1. **Replication:** Storing multiple copies of data or services across nodes.
2. **Redundancy:** Adding extra components to handle failures.
3. **Check pointing:** Periodically saving the system state to recover from failures.
4. **Recovery:** Restoring the system to a consistent state after a failure.

- ✧ **Transient Faults:** Transient Faults are the type of faults that occur once and then disappear. These types of faults do not harm the system to a great extent but are very difficult to find or locate. Processor fault is an example of transient fault.
- ✧ **Intermittent Faults:** Intermittent Faults are the type of faults that comes again and again. Such as once the fault occurs it vanishes upon itself and then reappears again. An example of intermittent fault is when the working computer hangs up.

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- ✧ **Permanent Faults:** Permanent Faults are the type of faults that remains in the system until the component is replaced by another. These types of faults can cause very severe damage to the system but are easy to identify. A burnt-out chip is an example of a permanent Fault.

- ✧ Fault Tolerance is required in order to provide below four features.
- ✧ **Availability:** Availability is defined as the property where the system is readily available for its use at any time.
- ✧ **Reliability:** Reliability is defined as the property where the system can work continuously without any failure.
- ✧ **Safety:** Safety is defined as the property where the system can remain safe from unauthorized access even if any failure occurs.
- ✧ **Maintainability:** Maintainability is defined as the property states that how easily and fastly the failed node or system can be repaired.



1. Fault Detection

- ✧ Fault Detection is the first phase where the system is monitored continuously. The outcomes are being compared with the expected output. During monitoring if any faults are identified they are being notified. These faults can occur due to various reasons such as hardware failure, network failure, and software issues. The main aim of the first phase is to detect these faults as soon as they occur so that the work being assigned will not be delayed.

2. Fault Diagnosis

- ✧ Fault diagnosis is the process where the fault that is identified in the first phase will be diagnosed properly in order to get the root cause and possible nature of the faults. Fault diagnosis can be done manually by the administrator or by using automated Techniques in order to solve the fault and perform the given task.

3. Evidence Generation

- ✧ Evidence generation is defined as the process where the report of the fault is prepared based on the diagnosis done in an earlier phase. This report involves the details of the causes of the fault, the nature of faults, the solutions that can be used for fixing, and other alternatives and preventions that need to be considered.

4. Assessment

- ✧ Assessment is the process where the damages caused by the faults are analyzed. It can be determined with the help of messages that are being passed from the component that has encountered the fault. Based on the assessment further decisions are made.

5. Recovery

- ✧ Recovery is the process where the aim is to make the system fault free. It is the step to make the system fault free and restore it to state forward recovery and backup recovery. Some of the common recovery techniques such as reconfiguration and resynchronization can be used.